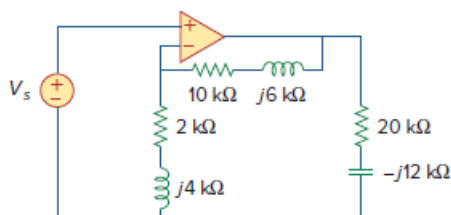


Problem

For the op amp circuit in Fig. 11.41, $V_s = 2\angle 30^\circ \text{ V}$. Find the average power absorbed by the 20-k Ω resistor.

Figure 11.41:



Step-by-step solution

Step 1 of 2

Refer to Figure 11.41 in the text book.

Write the expression for the output of the non-inverting operational amplifier.

$$V_o = \left(1 + \frac{Z_f}{Z_1} \right) V_s$$

Substitute $10 \times 10^3 + j(6 \times 10^3)$ for Z_f , $2 \times 10^3 + j(4 \times 10^3)$ for Z_1 and $2\angle 30^\circ \text{ V}$ for V_s .

$$\begin{aligned} V_o &= \left(1 + \frac{10 \times 10^3 + j(6 \times 10^3)}{2 \times 10^3 + j(4 \times 10^3)} \right) (2\angle 30^\circ) \\ &= \left(1 + \frac{10 + j6}{2 + j4} \right) (2\angle 30^\circ) \\ &= \left(\frac{16}{5} - j\frac{7}{5} \right) (2\angle 30^\circ) \\ &= 6.9857\angle 6.3706^\circ \text{ V} \end{aligned}$$

Step 2 of 2

Determine the output current I_o .

$$\begin{aligned} I_o &= \frac{V_o}{20 \times 10^3 - j(12 \times 10^3)} \\ &= \frac{6.9857\angle 6.3706^\circ}{20 \times 10^3 - j(12 \times 10^3)} \\ &= 2.995 \times 10^{-4} \angle 37.33^\circ \text{ A} \end{aligned}$$

Determine average power absorbed by the 20 k Ω resistor.

$$\begin{aligned} P &= \frac{1}{2} |I_o|^2 R \\ &= \frac{1}{2} (2.995 \times 10^{-4})^2 (20 \times 10^3) \\ &= 8.97 \times 10^{-4} \\ &= 897 \mu\text{W} \end{aligned}$$

Thus, the average power absorbed by the 20 k Ω resistor is $897 \mu\text{W}$.

